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Hamstring-to-quadriceps fatigue ratio offers new and different muscle function information than the conventional non-fatigued ratio

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Commonly used injury risk prediction tests such as the hamstring-to-quadriceps (H:Q) strength ratio appear to be poor predictors of non-contact injury. However, these tests are typically performed in a non-fatigued state, despite accumulated fatigue being an important risk factor for both hamstring strain (HS) and anterior cruciate ligament (ACL) injuries in professional soccer players. After the effect of different H:Q calculation methods were compared and contrasted, the influence of neuromuscular fatigue on the H:Q strength ratio and the association between fatigued and non-fatigued ratio scores were examined. Thirty-five professional soccer players performed a 30-repetition isokinetic fatigue test protocol. Peak knee joint moments were computed for each repetition, and the H:Q conventional ratio ($H:Q_{CR}$) was calculated using several different, previously published, methods. Knee extensor and flexor moments were statistically decreased by the sixth repetition and continued to decrease until the end of the protocol. However, the H:Q ratio was statistically decreased at the end of the test due to a significant reduction in knee flexor moment (correlation between change in knee flexor moment and change in H:Q, $r \approx .80$; $P < .01$). Moreover, H:Q measured in fatigue (ie, $H:Q_{Fatigue}$) at the end of the test was greater than $H:Q_{CR}$ (1.25-1.38 vs 0.70, $P < .01$), these variables were weakly correlated ($r = .39$, $P = .02$), and subject rankings within the cohort based on $H:Q_{CR}$ and $H:Q_{Fatigue}$ were different ($r_s = 0.25$, $P = .15$). The present data suggest that H:Q ratio measurement during a fatiguing test ($H:Q_{Fatigue}$) provides different outcomes to the traditional $H:Q_{CR}$. The observed significant hamstring fatigue and the difference, and weak correlation, between $H:Q_{CR}$ and $H:Q_{Fatigue}$ indicate that useful information might be obtained with respect to the prediction of HS and ACL injury risk. The potential predictive value of $H:Q_{Fatigue}$ warrants validation in future prospective trials.

KEYWORDS

anterior cruciate ligament and hamstring strain injuries, fatigue test, hamstring-to-quadriceps ratio, isokinetic, neuromuscular fatigue, soccer, test interpretation

1 | INTRODUCTION

The physical demands of match play soccer, including the performance of explosive acceleration and deceleration movements, often with concomitant changes in direction, have been shown to expose athletes to considerable hamstring strain (HS) and anterior cruciate ligament (ACL) injury risk.[1,2] Given the substantial time and financial costs

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associated with injury, the high rate of re-injury, and potential for ongoing health-related problems for the athlete,[3-5] tests aiming to predict injury likelihood (risk) and help with return-to-sport decisions are of great practical importance.

However, despite the prevalent use of some screening tests, recent data have shown that both HS and ACL (re)injury rates have not changed, or have even increased, in recent years,[6,7] possibly indicating an ineffectiveness of such tests to predict (re)injury. As an example, hamstring muscle force production in relation to the quadriceps is thought to be a strong predictor of lower limb injuries in soccer, including HS and ACL injury.[8-11] This concept is believed to have come to prominence early in the 1950s[9,12] and was subsequently popularized in the 1980s by the studies of Yasuda and Sasaki,[13] Solomonow et al.,[14] and Baratta et al.,[8] who highlighted the role of hamstring co-activation and hamstring-ACL synergism in counteracting anterior tibial translation and rotation during forceful knee extensions, as well as its potential injury preventive role.[8,13,14] Interestingly, the hamstring-to-quadriceps (H:Q) strength ratio is currently ranked as one of the five most common tests performed to quantify injury risk.[11,15] However, a recent large prospective trial revealed that the H:Q strength ratio was not predictive of future HS in 614 elite soccer players across four competitive seasons.[16] The conventional ratio ($H:Q_{CR}$), which is an outcome derived from a test usually performed using slow velocity, alternating knee extension and flexion concentric contractions, has sometimes been criticized on the basis that hamstring injuries occur during the eccentric phase of running stride when the knee joint angular velocity is high.[12,17] Yet these authors[16] also observed no predictive potential of the test when performed during eccentric actions or at higher joint angular velocities (ie, at both $60°\cdot s^{-1}$ and $300°\cdot s^{-1}$), which calls into question the predictive potential of several versions of the test. Therefore, alternative tests need to be developed, or the H:Q ratio may need to be tested under different muscle contraction conditions to have a greater predictive potential.

One factor that appears to strongly influence injury risk is the muscle fatigue developed during a match and/or accumulated during a season.[2,11,15] In fact, the highest number of HS injuries is observed in the final 15 minutes of each half of a soccer match,[2] and experimental work shows remarkable alterations in ACL injury risk markers (eg, altered mechanical and sensory function related to knee stability) after fatiguing exercise.[18] Therefore, one possible problem with current versions of the H:Q ratio test is that they have been performed under non-fatigued conditions, even though the H:Q ratio was previously demonstrated to be sensitive to experimentally imposed muscular fatigue.[19] For a "fatigued" H:Q test to be successful, it would clearly have to provide a different H:Q score as well as rank individuals differently within a cohort (or when compared to the population of H:Q ratios calculated and published previously). If this was the case, it would indicate that the ratio measured under fatigue conditions ($H:Q_{Fatigue}$) offers different information and therefore would be worth testing further for its potential to predict injury or the successful return to sport after (re)injury in future prospective trials. Therefore, the first purpose of this study was to complete an initial-phase trial to test whether the $H:Q_{Fatigue}$ ratio offers different information to the conventional (non-fatigued, $H:Q_{CR}$) ratio when measured under controlled, fatigue-inducing test conditions.

However, any H:Q test utilizing data collected during fatigue will first require estimation of the contractile decrement induced by the muscle work completed. A fatigue index (FI) is already often computed to discriminate players based on their susceptibility to fatigue.[20] It is commonly calculated as the torque produced in the last repetitions of a fatiguing test as a ratio of those produced in the first repetitions and is mainly obtained for knee extension.[21,22] However, there are notable differences between studies in the methods used to compute the FI. Researchers, as well as the software systems that accompany testing equipment such as isokinetic dynamometers (eg, Cybex NORM, Humac v. 12.17.0), have calculated FI as the percent decrease in the averaged peak moment over the last three repetitions relative to the averaged peak moment over the first three repetitions,[20,22] although the average of the first five and the last five repetitions has also been used in research.[23] These methods may be somewhat problematic due to the intrasubject variability of initiation and maintenance of force production during the test; that is, some subjects increase their torque production over the first repetitions, while others show immediate decline. It is currently unclear whether the choice of calculation method influences the fatigue index calculation, or indeed the conclusions drawn from the test results. Given that the method chosen to calculate torques at the start and end of a fatiguing test will also influence H:Q measured under conditions of fatigue ($H:Q_{Fatigue}$), and thus the conclusion drawn from this test, there is a need to examine the effect of using different calculations on both FI and $H:Q_{Fatigue}$ to determine the extent to which the calculation method influences the test results. Thus, after determining the "ideal method" for calculating FI in this study, we then compare and correlate $H:Q_{Fatigue}$ with $H:Q_{CR}$ to test for differences between these outcomes.

Specifically, the purpose of this study was to (a) test whether the H:Q ratio measured under conditions of fatigue provides a different score and is uncorrelated with $H:Q_{CR}$; (b) determine whether it ranks individuals differently when compared to the population of $H:Q_{CR}$ (non-fatigued) ratios; and (c) examine the strength of relationships between changes in $H:Q_{CR}$ and the FI developed in knee flexor and extensor muscle groups. Finally, because the choice of determining maximum joint moment influences the fatigue index calculation, we tested whether the use of different calculation methods affected the $H:Q_{Fatigue}$.

2 | METHODS

2.1 | Subjects

Thirty-five elite professional soccer players (26.0±4.6 years, 74.0±8.1 kg, and 178.0±6.4 cm) from Brazilian south state and national first division league clubs volunteered for this study. An informed consent form based on Declaration of Helsinki ethical principles for medical research involving human subjects was read and signed prior to testing. The subjects were asymptomatic and free of musculo-skeletal knee injuries, participated in training sessions on average six times per week during the preparation phase, and were familiar with the extension and flexion isokinetic tests (all had performed them in previous seasons). All tests were performed early in the preseason as part of the sport club's screening procedure, and the variables were assessed in the dominant leg by adopting a criterion of preferred kicking leg according to the participants. The regional research ethical committee approved this study.

2.2 | Isokinetic testing

The test procedures were completed as detailed in previous studies.[20,24] Briefly, maximal knee extensor and flexor peak moments were measured during a fatigue test in a Cybex Norm isokinetic dynamometer (Ronkonkoma, NY, USA). A 5-minute no-load warm-up was performed on a cycle ergometer at a set cadence of 75 rpm (Movement Technology, BM2700, São Paulo, Brazil) before the subjects were seated on the dynamometer with straps across their thigh (test leg), hips, and chest to minimize extraneous movement. The lateral epicondyle of the tested knee was aligned with the dynamometer's axis of rotation, and the machine's lever arm was attached to the lower leg 2 cm above the lateral malleolus of the ankle. The knee moment of the preferred limb (ie, kicking limb) was measured through a range of motion from 90° (knee flexion) to 0° (full knee extension). Prior to testing, the subjects performed a specific submaximal isokinetic warm-up of 10 extension-flexion repetitions at $120°\cdot s^{-1}$ and five repetitions at $300°\cdot s^{-1}$ for familiarization purposes. The subjects were instructed to perform contractions at a self-selected submaximal joint moment (perceived effort) in the first three repetitions and then to increase their force in the last two repetitions without reaching maximal perceived force production. Note that familiarization sessions prior to testing were not performed as (a) such familiarization was not provided in previous prospective trials,[16] and (b) players had experience with the testing in previous seasons and would not have likely been given additional practice time in their clubs.

The fatigue test consisted of 30 continuous repetitions of knee extensor and flexor concentric contractions at $300°\cdot s^{-1}$. This number of repetitions was chosen based on results from previous studies demonstrating significant reductions in knee flexion and extension torque after a similar fatiguing test in professional soccer players,[20] and that a minimum of 25 repetitions is needed to accurately determine the decrease in H:Q ratio.[25] Additionally, this speed was chosen based on the knowledge that HS and ACL injuries typically occur at high movement speeds,[1,17] and also because the fatigue index is often measured at high speeds.[20] Therefore, no inference will be made to H:Q test performed at slower speeds. The participants were instructed to start the test with maximal effort "without saving force for the end of the test" by extending and flexing as hard and as fast as possible while verbal encouragement was given during the test. Raw data were exported from the dynamometer's software program (HUMAC V. 12.17.0, Stoughton, MA, USA) to a personal computer after gravity correction and analyzed as described below.

2.3 | Data analysis based on frequency of the repetition in which maximal joint moment (Mom_{max}) was produced

A descriptive analysis revealed that approximately 80% of subjects produced their maximal knee extension and flexion moments between the 1st and the 5th repetitions, with approximately 70% producing maximal knee extension and flexion moments between the 2nd and 5th repetitions and ~63% producing it between the 2nd and 4th repetitions. This ~7% difference is explained by some subjects generating greater moments in the 1st or the 5th repetition compared that reached between the 2nd and 4th repetitions. The within-subject mean variances (%) between these repetitions were 3.1% and 4.6% for knee extension moment and 7.6% and 5.0% for knee flexion moment for the 1st and 5th repetitions, respectively. The mean variances were only computed for the subjects who produced greater moments in the 1st or 5th repetition compared to the mean moment between the 2nd, 3rd, and 4th repetitions and were calculated as the variation between the torques produced in the 1st or 5th repetitions and the mean torque value between the 2nd, 3rd, and 4th repetitions. Therefore, the mean value of the 2nd, 3rd, and 4th repetitions (ie, *maximum* or maximal joint moment) was considered to represent the maximal joint moment produced and was thus used in all comparisons made in this study (joint moment and $H:Q_{CR}$ changes, FI calculation and correlation between $H:Q_{CR}$ and FI changes), unless otherwise stated. For convenience, "*maximum*" or "maximal joint moment/torque" (Mom_{max}) will be referred to as the mean value of the 2nd, 3rd, and 4th repetitions, while "peak moment/torque" refers to the peak joint moment ($PMom_{rep}$, where "rep" is the specific repetition number) produced in a specific repetition.

TABLE 1 Equations used to calculate the different hamstring-to-quadriceps ratios. Maximum joint moment and mean/average peak moment are often used to compute the strength ratio, and the following equations for the fatigue ratio (Fatigue $Ratio_{(repetition)}$) and Mean Fatigue Ratio ($Mean_{(repetitions)}$Fatigue Ratio) are proposed in the present study as new variables that might be of functional relevance

Strength Ratio (abbreviation)	Description	Equation
Fatigue $Ratio_{(28,29,30)}$	Fatigue ratio calculated using the 28th, 29th, and 30th repetitions (individual comparisons)	Knee Flexor $FI_{28,29,30}$/Knee Extensor $FI_{28,29,30}$
$Mean_{(28\text{-}30)}$Fatigue Ratio	Mean FI value of the last three repetitions	Knee Flexor $FI_{28\text{-}30}$/Knee Extensor $FI_{28\text{-}30}$
M-H:Q_{CR}	Maximal H:Q conventional ratio utilizing the maximum joint moment (Mom_{max})	Knee Flexor Mom_{max}/Knee Extensor Mom_{max}
$P_{1\text{-}3}$-H:Q_{CR}	Mean joint moment ratio of the first three repetitions	Knee Flexor $PM_{1\text{-}3}$/Knee Extensor $PM_{1\text{-}3}$
$P_{1\text{-}5}$-H:Q_{CR}	Mean joint moment ratio of the first five repetitions	Knee Flexor $PM_{1\text{-}5}$/Knee Extensor $PM_{1\text{-}5}$
P_1-H:Q_{CR}	Joint moment of the first repetition	Knee Flexor PM_1/Knee Extensor PM_1

2.4 | Peak moment change during the fatigue test

To identify the changes in knee extension and flexion moments during the fatigue test, the peak moment achieved in each repetition (ie, all 30 repetitions) was compared to the maximal joint moment.

2.5 | Hamstring-to-quadriceps conventional ratio (H:Q_{CR})

The maximal H:Q conventional ratio (M-H:Q_{CR}) was calculated by dividing the knee flexor *maximum* joint moment by the knee extensor *maximum* joint moment, while the peak moment H:Q conventional ratio (P-H:Q_{CR}) was calculated by dividing the mean knee flexor concentric *peak* moment over five repetitions by the mean knee extensor concentric *peak* moment measured over five repetitions throughout the test (ie, 1st to 5th, 6th to 10th, 11th to 15th; see Table 1). It should be noted that most researchers calculate the average of the peak joint moments for the first three repetitions[23] or the "best" peak joint moment in a test consisting of three to five reciprocal contractions.[16,26,27] It is difficult to compare methods used to calculate the *maximum* joint moment due to the lack of similarity between this study and previous studies (ie, 3 or 5 vs 30 repetitions). The most similar approach[23] calculated the H:Q_{CR} change at each fifth repetition over 50 repetitions. These authors considered the maximum H:Q_{CR} to be the mean torque of three repetitions, which may not be the highest value as some subjects do not reach the highest torque in the initial repetitions of the test (as observed in this study). Therefore, we opted to calculate H:Q_{CR} using the same method as for FI (see Table 2), and the decrease in H:Q_{CR} is graphically represented in a window of five repetitions for simplicity.

TABLE 2 Equations used to calculate the knee extensor and flexor fatigue index (FI)

FI method	Equation
$A_{28,29,30}$	$[(Mom_{max}\text{-}P_{28,29,30}/Mom_{max})\times100]$
B	$[(PM_{1\text{-}3}\text{-}P_{28\text{-}30}/PM_{1\text{-}3})\times100]$
C	$[(PM_{1\text{-}5}\text{-}P_{26\text{-}30}/PM_{1\text{-}5})\times100]$

Mom_{max}=maximal joint moment; $PM_{28,29,30}$=peak moment of the 28-30 repetitions (individual comparisons); $PM_{1\text{-}3}$=mean of the joint moment in repetitions 1-3; $PM_{28\text{-}30}$=mean joint moment in repetitions 28-30; $PM_{1\text{-}5}$=mean joint moment in repetitions 1-5, $PM_{26\text{-}30}$=mean joint moment in repetitions 26-30.

To compute changes in the knee flexion-to-extension strength ratio, P-H:Q_{CR} was compared over different contraction ranges to M-H:Q_{CR}. Moreover, the average H:Q_{CR} for the first three and five repetitions was computed to enable a comparison of the different methods for calculating the strength ratio (Table 1).

2.6 | Hamstring-to-quadriceps fatigue ratio (H:$Q_{Fatigue}$)

H:$Q_{Fatigue}$, which is also referred to as the Fatigue $Ratio_{rep}$ (where "rep" is the specific repetition number), was calculated using the FI individually computed for each of the last three repetitions and also averaged over the last three repetitions ($Mean_{(28\text{-}30)}$Fatigue Ratio). Table 1 lists the equations used to calculate these variables.

2.7 | Fatigue index (FI)

FI was calculated using methods described previously[20,22,23] as well as using an alternative method where the percent decrease in the peak joint moment over the last three

repetitions was compared to the maximal joint moment; this method will be referred to as Method A and is a new method proposed in this study (see Table 2). Moreover, the percent decrease in the average peak moment averaged over the last three repetitions relative to the average of the first three repetitions was also calculated as Method B.[20,22] Finally, the percent decrease in the average of the peak moments achieved in the last five repetitions relative to the average of the first five repetitions was computed as Method C. The outcomes provided from these methods were subsequently compared.

2.8 | Statistical analysis

Descriptive data are shown as mean±standard deviation (mean±SD), and the normality of all values was verified by the Shapiro-Wilk test. Data without normal distribution were log-transformed before parametric statistical analysis. When log-transformed data did not meet the criterion of normal distribution, Wilcoxon signed-rank test was performed. For normally distributed data, paired-samples *t* tests were used followed by false discovery rate analysis to correct for type I errors that may result from multiple *t* tests[28] to compare (a) concentric knee extensor and knee flexor moments, (b) the $H:Q_{CR}$ during the fatiguing test repetitions, and (c) the different methods of $H:Q_{CR}$ and fatigue index calculations. In addition, standardized effect sizes and their respective 95% confidence intervals were calculated as they are not substantially affected by sample size and data variability, and do not give a dichotomous answer (ie, significant or not significant).[29] Pearson's product-moment correlations (*r*) were computed for parametric data to quantify the strength of relationships between the percentage changes in $H:Q_{CR}$ and fatigue indices in both muscle groups, and between Fatigue $Ratio_{30}$ and $H:Q_{CR}$, while Spearman's rank-order correlations were computed for nonparametric data (ie, Fatigue $Ratio_{28}$ and Fatigue $Ratio_{30}$ and $H:Q_{CR}$). In addition, individual changes were graphically inspected by raw and relative (ie, ranked against others within the subject sample) scores of the $H:Q_{CR}$ and $H:Q_{Fatigue}$ to verify any possible relationships between the methods. All data were analyzed using SPSS statistical software (version 20.0; SPSS, Chicago, IL, USA) with a level of significance set a priori at $\alpha=0.05$.

3 | RESULTS

3.1 | Change in absolute peak moment during the fatiguing test

As shown in Figure 1, knee extensor Mom_{max} was significantly greater than that produced on average during the 1st, 2nd, and 3rd repetitions ($P=.045$, .037, and .033, respectively) but not different to that produced in the 4th and the 5th repetitions ($P>.05$), indicating that peak moments were most often produced in repetitions 2-4. It should be noted that a large effect size was calculated for the comparison of Mom_{max} vs torque produced in the 1st repetition [$d=1.16$ (-1.66 to -0.65)], while small effect sizes were calculated for Mom_{max} vs the 2nd and 3rd repetitions ($0.005\leq d\geq 0.086$). However, knee flexor Mom_{max} was significantly greater than that produced on average during repetition 1 [$P<.01$; $d=0.58$ (-1.1 to -0.1)] but not significantly different to repetitions 2-5 ($P>.05$; $0.03<d>0.15$). Thus, the joint moment in repetition 1 was not an accurate indication of the maximum joint moment achievable during the test (Mom_{max} mean values: 130.2 and 91.4 N·m; peak mean moment in the first repetition: 107.8 and 81.3 N·m, for both knee extensor and flexor values, respectively). Statistical analysis also revealed that both the knee extensor and flexor moments were significantly reduced by the 6th repetition ($P=.045$ and 0.037, respectively) and decreased further in each successive repetition of the protocol ($P<.001$).

3.2 | Comparisons between $H:Q_{CR}$ and $H:Q_{Fatigue}$ methods

Data that were log-transformed were compared by paired-samples *t* tests as follows: $M\text{-}H:Q_{CR}$ vs Fatigue $Ratio_{28}$, $M\text{-}H:Q_{CR}$ vs Fatigue $Ratio_{29}$, $M\text{-}H:Q_{CR}$ vs $P_{1\text{-}3}\text{-}H:Q_{CR}$, and $Mean_{(28\text{-}30)}$Fatigue Ratio vs Fatigue $Ratio_{28}$, and $Mean_{(28\text{-}30)}$Fatigue Ratio vs Fatigue $Ratio_{29}$. Comparisons between Fatigue $Ratio_{28}$ vs $P_{28}\text{-}H:Q_{CR}$ and Fatigue $Ratio_{29}$ vs $P_{29}\text{-}H:Q_{CR}$ did not meet normal distribution even after log-transformation and were thus compared by Wilcoxon signed-ranked test. All data are presented as absolute values.

Statistical analysis revealed that the $M\text{-}H:Q_{CR}$ was significantly lower than the peak moment conventional ratio obtained in the 1st repetition ($P_1\text{-}H:Q_{CR}$) ($P=.038$), as shown in Table 3. These statistical differences also demonstrate that the results of the statistical analyses were different when the changes in H:Q ratio were compared to the 1st repetition than when comparing to $M\text{-}H:Q_{CR}$. However, calculating this variable using the average between the moments produced in the first three or first five repetitions reduced this difference; statistical differences were observed between $M\text{-}H:Q_{CR}$ and $P_{1\text{-}3}\text{-}H:Q_{CR}$ ($P<.001$) but not $P_{1\text{-}5}\text{-}H:Q_{CR}$ ($P=.09$). These results are presented in the lower part of Table 3.

$H:Q_{Fatigue}$ calculated using the moment developed in repetitions 28-30 was significantly greater than $M\text{-}H:Q_{CR}$ (although statistical analysis revealed no significant differences between $M\text{-}H:Q_{CR}$ and Fatigue $Ratio_{29}$, large effect sizes between methods were found and indicated a large effect of calculation method). $H:Q_{Fatigue}$ values measured using repetitions 28-30 were significantly larger than measured using $P\text{-}H:Q_{CR}$ repetitions 28-30. Collectively, these results demonstrate that different values are observed when the ratio is calculated using different methods.

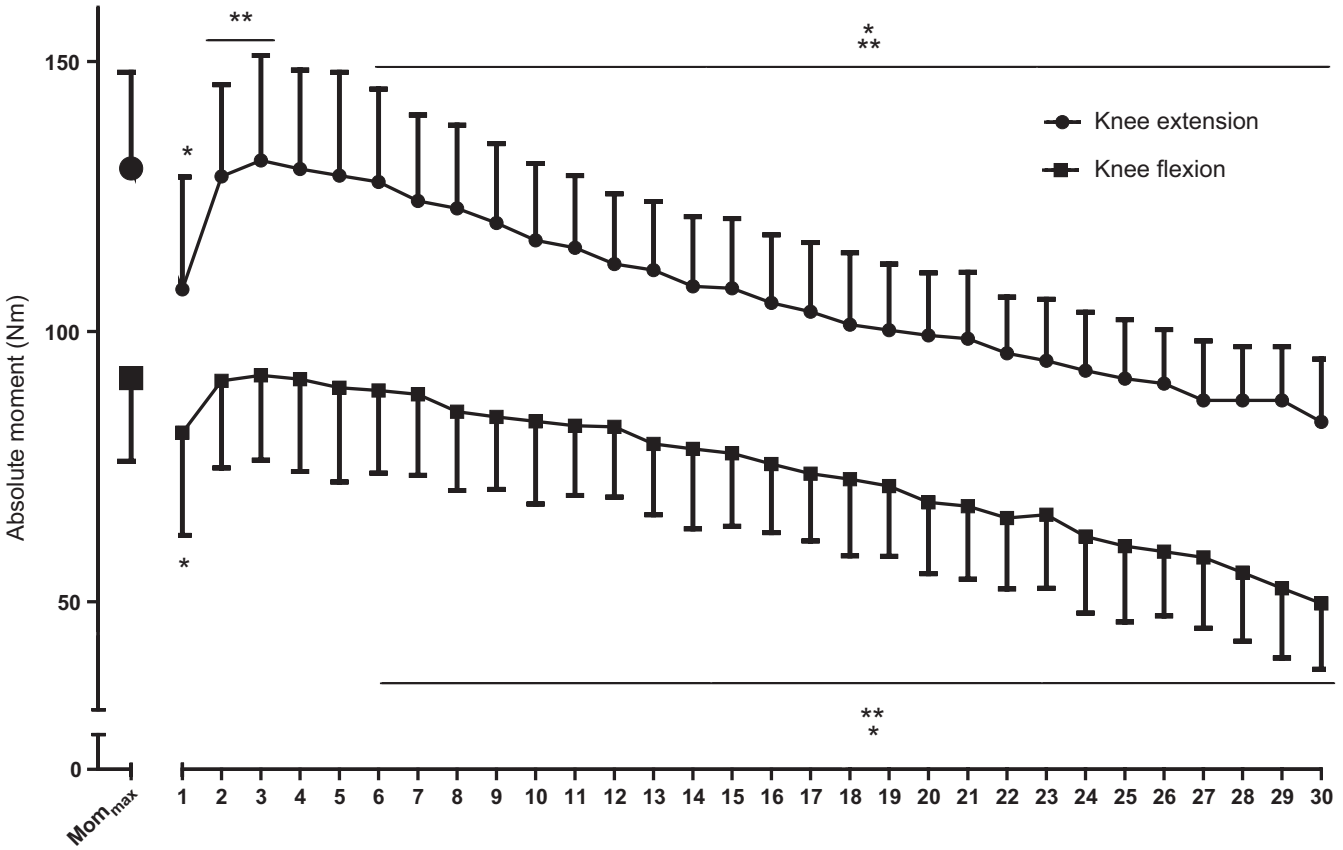

FIGURE 1 Knee extensor and flexor moments during the fatigue test. Knee extensor moment in the 1st and 2nd repetitions were significantly lower than the maximal joint moment (Mom_{max}, mean from the 2nd to the 4th repetitions, indicated as the large filled square and circle), and joint moment in repetition 3 was significantly higher than Mom_{max} [although a very small effect size was observed, d=0.08 (−0.38 to 0.54)], while the joint moment produced in repetition 1 was significantly lower than the maximal joint flexor moment. Both extensor and flexor moments decreased by the 6th repetition and remained significantly decreased to the end of the protocol when compared to the Mom_{max}. *Significant difference from Mom_{max} to P<.05, ** significant difference from Mom_{max} to P<.01

When comparing the different methods for calculating $H:Q_{Fatigue}$, statistical analysis revealed the $Mean_{(28\text{-}30)}$Fatigue Ratio to be significantly greater than the Fatigue $Ratio_{28}$ (P=.007) and significantly lower than the Fatigue $Ratio_{29}$ (P=.036), but not different to Fatigue $Ratio_{30}$. Collectively, these results indicate that computing $H:Q_{Fatigue}$ for each repetition may be statistically different to calculating the average ratio across the last three repetitions. These outcomes are presented in the upper part of the Table 3.

3.3 | Association between fatigue ratio ($H:Q_{Fatigue}$) and maximal hamstring-to-quadriceps conventional ratio ($M\text{-}H:Q_{CR}$)

In relation to the main aim of this study, Spearman's correlations demonstrated no significant relationship between $M\text{-}H:Q_{CR}$ and Fatigue $Ratio_{28}$ (r_s=0.27, P=.11) or Fatigue $Ratio_{29}$ r_s=0.25, P=.15). Additionally, Pearson's product-moment correlations showed a weak positive correlation (r=.39, P=.02) between $M\text{-}H:Q_{CR}$ and Fatigue $Ratio_{30}$. Overall, these findings indicate only a weak (or no) correlation between $H:Q_{Fatigue}$ and $H:Q_{CR}$, so the ranking of individuals in the sample differed depending on the variable calculated. In addition, visual inspection of individual changes revealed no standard pattern of change when raw and rank scores were graphically represented (see example in Figure 2). Thus, the subjects who scored best or worst in $H:Q_{CR}$ did not necessarily maintain this position using the $H:Q_{Fatigue}$ method.

3.4 | Change in absolute hamstring-to-quadriceps conventional ratio and fatigue index developed between knee flexion and extension

There were significantly greater reductions in $P_{26\text{-}30}\text{-}H:Q_{CR}$ compared to $M\text{-}H:Q_{CR}$ [P<.001; d=0.70 (−0.21 to −1.18)], as shown in Figure 3A. It should be noted that the statistical analysis revealed a significant decline only in repetitions 28-30 (P<.01 for all; data not shown) when analyzing the individual $P\text{-}H:Q_{CR}$ changes across repetitions. These reductions were also accompanied by a significantly greater decrease in the knee flexor than knee extensor moment (ie, FI; Figure 3B).

TABLE 3 Comparisons between different hamstring-to-quadriceps conventional ratio ($H:Q_{CR}$) and fatigue ratio ($H:Q_{Fatigue}$) calculations (see Table 1 for calculation definitions)

Comparisons	Mean±SD		Effect Size (95% CI)
	Fatigue Ratio	H:Q	
Fatigue $Ratio_{28}$ vs $M\text{-}H:Q_{CR}$	1.25±0.45	0.70±0.09**	1.69 (1.14 to 2.24)
Fatigue $Ratio_{29}$ vs $M\text{-}H:Q_{CR}$	1.38±0.48	0.70±0.09	1.96 (1.39 to 2.54)
Fatigue $Ratio_{30}$ vs $M\text{-}H:Q_{CR}$	1.33±0.38	0.70±0.09**	2.28 (1.67 to 2.88)
Fatigue $Ratio_{28}$ vs $P_{28}\text{-}H:Q_{CR}$	1.25±0.45	0.63±0.13**	1.87 (1.31 to 2.44)
Fatigue $Ratio_{29}$ vs $P_{29}\text{-}H:Q_{CR}$	1.38±0.48	0.60±0.12**	2.23 (1.63 to 2.83)
Fatigue $Ratio_{30}$ vs $P_{30}\text{-}H:Q_{CR}$	1.33±0.38	0.59±0.12**	2.63 (1.98 to 3.26)
$Mean_{(28\text{-}30)}$Fatigue Ratio vs Fatigue $Ratio_{28}$	1.31±0.39	1.25±0.45**	0.14 (−0.61 to 0.32)
$Mean_{(28\text{-}30)}$Fatigue Ratio vs Fatigue $Ratio_{29}$	1.31±0.39	1.38±0.48*	0.16 (−0.31 to 0.63)
$Mean_{(28\text{-}30)}$Fatigue Ratio vs Fatigue $Ratio_{30}$	1.31±0.39	1.33±0.38	0.05 (−0.4 to 0.5)
$M\text{-}H:Q_{CR}$ vs $P_{1\text{-}3}\text{-}H:Q_{CR}$	0.70±0.09	0.72±0.1*	0.21 (−0.26 to 0.68)
$M\text{-}H:Q_{CR}$ vs $P_{1\text{-}5}\text{-}H:Q_{CR}$	0.70±0.09	0.71±0.09	0.11 (−0.36 to 0.58)
$M\text{-}H:Q_{CR}$ vs $P_{1}\text{-}H:Q_{CR}$	0.70±0.09	0.77±0.18*	0.49 (0.02 to 0.97)

*Significant to $P<.05$; **Significant to $P<.01$.

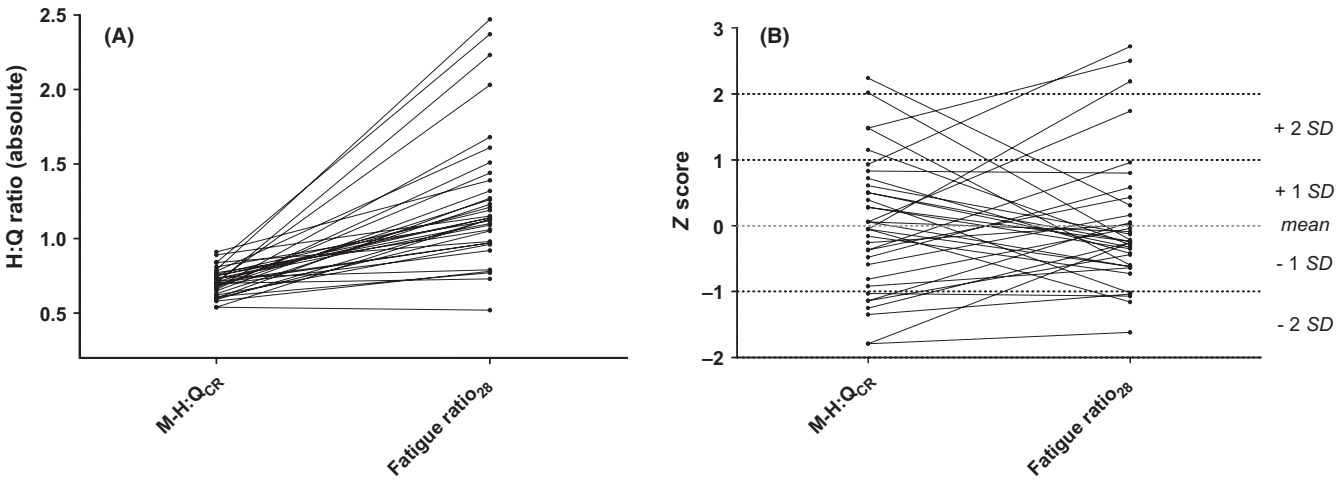

FIGURE 2 Example of individual changes in hamstring-to-quadriceps ratios ($M\text{-}H:Q_{CR}$ and Fatigue $Ratio_{28}$) demonstrated by raw (A) and relative (B) scores, showing nonstandard changes in both methods of score representation

3.5 | Differences observed between fatigue index (FI) methods

Overall differences were observed between Method A and Method B and between Method A and Method C (see Table 2 for method definitions) when used to compute the knee flexor and extensor FIs. It should be noted that Method B and Method C underestimate the true decline in moment in the last two repetitions, and thus, FI was subsequently also underestimated (as noted in the highest values observed in Method

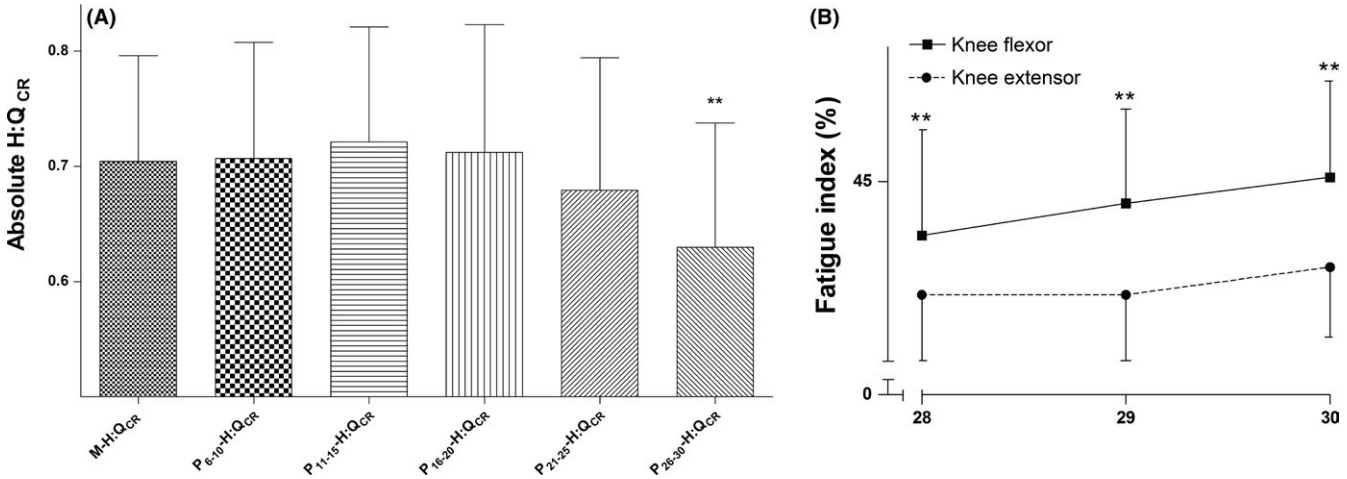

FIGURE 3 (A) The average hamstring-to-quadriceps conventional ratio calculated for every five repetitions was compared to M-H:Q_{CR}. A significant decline was observed in the last five repetitions (last column) compared to the M-H:Q_{CR}. (B) The fatigue index (FI) for the last three repetitions was significantly higher for FI calculated in the knee flexors compared to the knee extensors (see percentage differences in Table 4). **Significant to P<.01

TABLE 4 Comparisons between fatigue index (FI) calculation methods. Data are reported as mean±SD and ES (95% CI)

	FI Extensor			FI Flexor		
Method	**Mean±SD (%)**		**Effect Size (95% CI)**	**Mean±SD (%)**		**Effect Size (95% CI)**
A_{28} vs B	32.4±7.3	29.0±10.4	0.37 (−0.09 to 0.85)**	38.9±11.8	39.6±13.3	0.05 (−0.5 to 0.4)
A_{28} vs C	32.4±7.3	29.9±8.4	0.32 (−0.15 to 0.78)**	38.9±11.8	37.8±11.1	0.09 (−0.37 to 0.56)
A_{29} vs B	32.4±7.3	29.0±10.4	0.37 (−0.09 to 0.85)**	42.6±10.5	39.6±13.3	0.25 (−0.2 to 0.7)**
A_{29} vs C	32.4±7.3	29.9±8.4	0.32 (−0.15 to 0.78)**	42.6±10.5	37.8±11.1	0.44 (−0.03 to 0.9)**
A_{30} vs B	35.5±7.8	29.0±10.4	0.70 (0.2 to 1.19)**	45.5±10.7	39.6±13.3	0.48 (0.01 to 0.96)**
A_{30} vs C	35.5±7.8	29.9±8.4	0.69 (−1.17 to −0.2)**	45.5±10.7	37.8±11.1	0.71 (0.22 to 1.18)**
B vs C	29.0±10.4	29.9±8.4	0.09 (−0.37 to 0.56)	39.6±13.3	37.8±11.1	0.14 (−0.61 to 0.32)*
A_{30} vs A_{28}	35.5±7.8	32.4±7.3	0.41 (−0.88 to 0.06)**	45.5±10.7	38.9±11.8	0.59 (−1.06 to −0.10)**
A_{30} vs A_{29}	35.5±7.8	32.4±7.3	0.41 (−0.88 to 0.06)**	45.5±10.7	42.6±10.5	0.27 (−0.74 to 0.19)**
A_{28} vs A_{29}	32.4±7.3	32.4±7.3	—	38.9±11.8	42.6±10.5	0.33 (−0.80 to 0.14)**

*Significant to P<.05; **Significant to P<.01.

$A_{29,30}$ compared to the values in Method B and Method C). In addition, knee flexor FI calculated using Method B was significantly greater than when using Method C (P=.042), but was not significantly different for knee extensor FI (P=.137). Fatigue index comparisons are presented in Table 4.

3.6 | Relationship between changes in H:Q_{CR} and FI

Pearson's product-moment correlations computed between the reductions in H:Q_{CR} (percentage decline) and fatigue index from repetitions 28-30 were significant and positive for the knee flexor moments (r=.83, P<.001; r=.84, P<.001; and r=.80, P<.001, respectively), indicating that the subjects who had a greater (percent) decline in H:Q_{CR} also tended to demonstrate a greater fatigue in the knee flexors. Moreover, Pearson's correlation analysis revealed that subjects who had a smaller percent decline in the H:Q_{CR} also presented greater knee extensor fatigue in the 29th repetition (r=−.34, P=.03), but no significant correlations were observed for the changes in H:Q_{CR} or FI of the knee extensors in repetitions 28 and 30. These findings indicate that knee extensor FI had a weak (or no) effect on the reductions in H:Q_{CR} and that knee flexor fatigue may better explain the reductions in H:Q_{CR} during a fatigue test than knee extensor fatigue. These findings are graphically represented in Figure 4.

4 | DISCUSSION

The results of this study demonstrate that H:$Q_{Fatigue}$ (ranging 1.25-1.38 depending on the calculation method) was

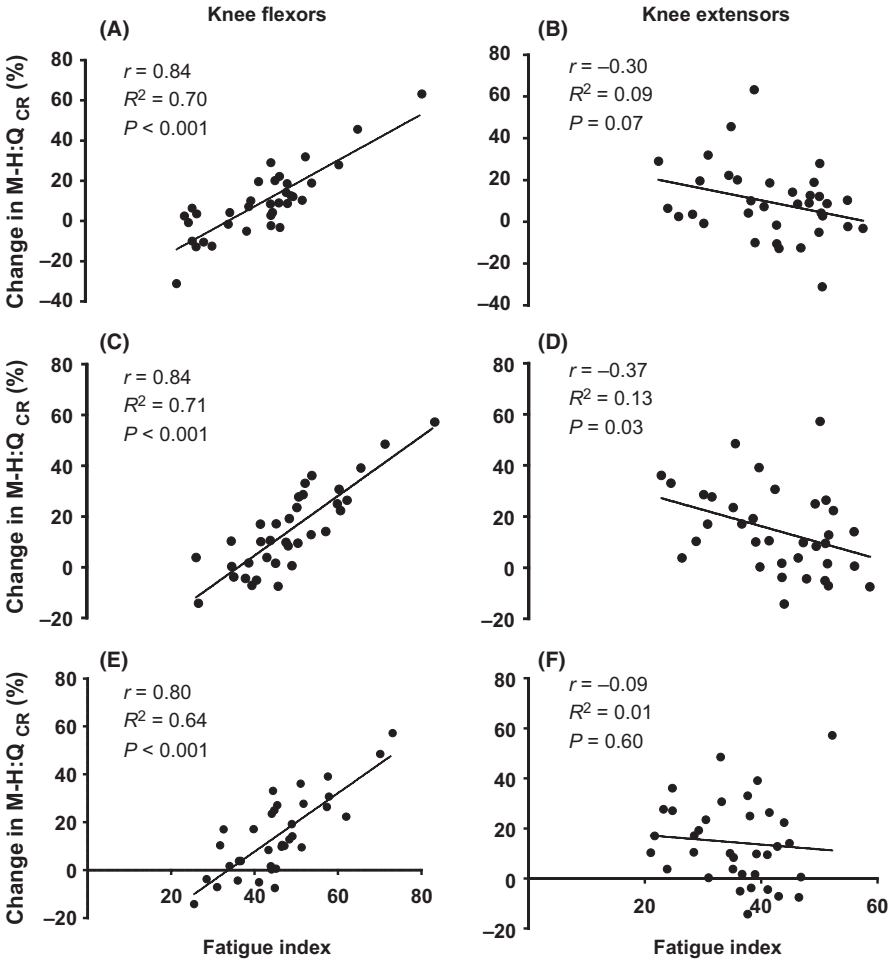

FIGURE 4 Relationships between changes in M-H:Q_{CR} and fatigue index (FI) of the knee flexors and extensors in repetitions 28-30 (A, C, and E—knee flexors; B, D, and F—knee extensors, respectively). A moderate-to-strong positive correlation was found for the changes in M-H:Q_{CR} and the FI in the knee flexors for repetitions 28 (r=.84), 29 (r=.84), and 30 (r=.80). The negative correlation observed between the changes in M-H:QCR and FI in the knee extensors was weak in the 29th repetition (r=−.37), and no significant relations were found for the 28th and 30th repetitions, indicating that knee flexor fatigue was moderately-to-strongly related to the decreases in M-H:Q_{CR}

significantly higher than the conventional (non-fatigued) maximal H:Q ratio (M-H:Q_{CR}=0.70; see Table 3) and was weakly correlated with H:Q_{CR} (r_s=0.25-0.27; r=.39). In fact, a different rank order of subjects' scores was depicted by the individual non-standardized rank changes between the H:$Q_{Fatigue}$ and M-H:Q_{CR} (Figure 2). Together, these results clearly indicate that (a) different conclusions would be drawn with respect to injury risk if a fatigue test was performed and (b) potentially different physiological effects must influence them. Several researchers have proposed that a conventional ratio below 0.66 and low levels of eccentric hamstring force capacity were potential risk factors for HS and/or ACL injury.[12,15] However, it should be kept in mind that soccer match play requires intermittent high-intensity tasks to be performed for a considerable time (~90 minutes) and thus is dependent upon the ability to develop maximal muscle force during sprinting, jumping, kicking, etc., while muscles become fatigued. Because maximal muscle force is evaluated in the H:Q strength ratio rather than the muscle endurance capacity, we speculate that H:$Q_{Fatigue}$ may be a better predictor of HS and ACL injury. An important next step, therefore, is to examine whether the information obtained using this fatigue-based test, that is, H:$Q_{Fatigue}$, can predict previous or subsequent HS and ACL injuries resulting from soccer match play during a season and to determine when players might safely and successfully return to sport after these injuries.[30]

In the present study, it was found that knee extension and flexion moments gradually decreased from the 6th repetition until the end of the fatiguing test. These findings are in partial agreement with those of Sangnier and Tourny-Chollet[23] who found knee extension and flexion moments to be decreased by the 15th repetition, although different angular velocities were utilized in their study ($180°\cdot s^{-1}$) compared to ours ($300°\cdot s^{-1}$). Regardless of these differences, the rate of decrease in knee joint moments may provide important information regarding an athlete's ability to maintain force in short-duration fatigue tasks (ie, the initial reduction in force capacity) as well as in "long-lasting" tasks when H:Q_{CR} is significantly reduced (ie, over the 28th to the 30th repetitions in the present study).

Interestingly, despite the simultaneous reduction in both knee extension and flexion moments, the H:Q_{CR} also showed a significant decline. The observed reduction in H:Q_{CR} in the last three repetitions of the fatigue test was accompanied by, and strongly correlated with, reductions in knee flexor moment, yet it was weakly (or not) correlated with

knee extension moment decline; that is, knee flexor fatigue appeared to explain the reductions in $H:Q_{CR}$ more than knee extensor fatigue. Moreover, the significantly greater fatigue index observed in the knee flexors compared to knee extensors may provide some indication as to why hamstring muscles are at greater risk of injury. Besides their muscle architectural complexity and multijoint function, the hamstrings also comprise a remarkable number of type II fibers and are less utilized in activities of daily living and are thus more susceptible to muscle fatigue as well muscle damage.[31-33] Although the mechanisms of HS and ACL injury are still under debate, it is known that neuromuscular fatigue is an important risk factor. For example, alterations in hamstring intermuscular coordination,[34,35] changes in hamstring muscle length during the late swing phase of running cycle,[36] and altered ACL-hamstring reflex associated with muscle stiffness regulation and knee stability[18] have been shown after fatiguing protocols. Collectively, these findings also speculatively suggest that the knee flexors might be trained to increase their force production and endurance capacities, and this may decrease one's susceptibility to HS and ACL injuries as a soccer match (or season) progresses.

Because several methods have been used to determine the *maximum* torque at the start and end of a fatiguing test,[20,22,23] and that the chosen method could potentially affect the H:Q strength ratio and FI, and therefore $H:Q_{Fatigue}$, several methods of $H:Q_{CR}$, $H:Q_{Fatigue}$, and FI calculation were tested in the present study. Regarding $H:Q_{CR}$, a significant difference was observed between the $H:Q_{CR}$ computed using the maximal joint moment and the average of the first three repetitions ($M\text{-}H:Q_{CR}$ vs $P_{1\text{-}3}\text{-}H:Q_{CR}$), but not compared to average of the first five repetitions ($M\text{-}H:Q_{CR}$ vs $P_{1\text{-}5}\text{-}H:Q_{CR}$). These findings may be explained by the significant difference and moderate-to-large effect sizes observed between the *maximum* joint moment (Mom_{max}) and the peak moment produced in the first repetition, which was sufficient to significantly affect $P_{1\text{-}3}\text{-}H:Q_{CR}$ but not the calculation of $H:Q_{CR}$. With respect to $H:Q_{Fatigue}$, we observed significant overall differences between $H:Q_{Fatigue}$ calculated for each repetition (individually calculated in the final three repetitions) and calculated using the average ratio across the last three repetitions (ie, $Mean_{(28\text{-}30)}$Fatigue Ratio), although small effect sizes were found; there is a need to test whether this difference in effect size affects the prediction of, or relationship with, injury incidence. Regarding FI methods, an overall difference was observed between Method A and Method B as well as Method A and Method C (see method descriptions in Table 2) when they were used to compute the knee flexor and extensor FIs. It should be noted that the classic method (Method B) proposed by Thorstensson and Karlsson,[22] and also the method (Method C) used by Sangnier and Tourny-Chollet,[23] underestimated the true decline in moment in the last two repetitions, and thus, the FI was underestimated; these findings are probably explained by the moderate-to-large effect sizes found between Mom_{max} and the peak torque in the first repetition and/or the variability in maintenance of force production in the final repetitions of the test, which supports our previous hypothesis. In addition, the frequency analysis of the repetition to which maximal moment was produced indicated that ~63% of subjects produced their maximal joint moment between the 2nd and 4th repetitions. These findings lend support to the notion that intrasubject variability of initiation and maintenance of force production during the test is considerably high, especially in the first repetition, and this can affect the calculation of FI as well as the changes in joint moment and H:Q ratio during the test. In addition, the knee flexor FI calculated using Method B was significantly greater than using Method C, but was not significantly different for knee extensor FI. These findings highlight significant and practically important differences in the selection of the method to calculate the FI, with Method A perhaps being more precise than Method B or Method C. However, the dynamometer angular velocity utilized in the present study was greater than that used in previous studies [$300°\cdot s^{-1}$ in the present study vs $180°\cdot s^{-1}$ used in Sangnier and Tourny-Chollet[23] and Thorstensson and Karlsson[22]]. The faster speed of $300°\cdot s^{-1}$ may speculatively exhaust the fast twitch fibers, that is, type II muscle fibers, to a greater extent than contractions at $180°\cdot s^{-1}$, which may partly explain the different results. Future studies are required to directly compare FI calculations and H:Q ratios (fatigue ratio and peak moment ratio) in both knee extensor and flexor muscles under conditions of different angular velocities. While it must be kept in mind that the best angular velocity to evaluate muscle knee imbalance is still debated (due to the conflicting results found in the literature[26,37]) it seems plausible that a relatively higher isokinetic velocity of testing mimics to a better extent the speed of muscle contraction during fast and explosive movements of soccer match play.

5 | LIMITATIONS

There are several points to consider with regard to the present research. First, professional soccer players were evaluated in this study at the beginning of the preseason when their physical preparation was commencing prior to competition. Their limited time for laboratory visits in this phase of the season did not allow for a test-retest reliability assessment, although the subjects were familiar with the tests due to screening evaluations being commonly performed in previous seasons. Future studies should determine the reliability of these outcomes measures, and particularly $H:Q_{Fatigue}$. Second, the decrease in eccentric hamstring muscle force could not be evaluated in the present professional players because of concerns relating to the muscle recovery time required and its

impact on training. It has been suggested that the capacity to maintain hamstring muscle eccentric force is associated with the (re)injury occurrence,[34,35] so this might be worth evaluating in the future. However, it should be noted that the evaluation of the capacity to maintain hamstring eccentric strength, as previously suggested,[35] is a test that exposes players to potentially significant hamstring muscle damage, and a long recovery time would be required.

6 | CONCLUSION

The present data suggest that the H:Q ratio measurement during a fatiguing, knee extension-flexion test ($H:Q_{Fatigue}$) may provide useful information with respect to the prediction of HS and ACL injury risk because $H:Q_{Fatigue}$ was significantly greater than the conventional (non-fatigued) maximal H:Q ratio ($M\text{-}H:Q_{CR}$), was weakly or not correlated with $M\text{-}H:Q_{CR}$, and presented a different rank order of subjects' scores depicted by the individual non-standardized rank changes between the $H:Q_{Fatigue}$ and $M\text{-}H:Q_{CR}$. The results revealed a greater muscle knee flexor compared to knee extensor muscle fatigue, and the strong correlations observed between knee flexor fatigue indices (FIs) and the changes in $H:Q_{CR}$ support the notion that hamstrings are susceptible to exceptional fatigue and may thus be at a greater injury risk. In addition, significant differences were observed between the $H:Q_{CR}$ computed by the maximal joint moment and the average between the first three repetitions, but not compared to the first five repetitions (although significant overall differences were observed between the individual $H:Q_{Fatigue}$ calculations over each three final repetitions compared to the average $H:Q_{Fatigue}$). Importantly, the methods used to compute FIs proposed in the present study demonstrated significant overall differences to those methods previously used to calculate them. Based on these results, we recommend the use of maximum joint moment (Method A—average between repetitions 2-4) to reduce variability and increase the accuracy of fatigue index estimation, and thus $H:Q_{Fatigue}$ accuracy. The present results provide support for the use of $H:Q_{Fatigue}$ in future prospective trials.

7 | PERSPECTIVES

Lower hamstring muscle force in relation to quadriceps (ie, hamstring-to-quadriceps ratio) has been shown to be a weak predictor of hamstring strain and ACL injury.[16,38] This may be, in part, because these injuries mostly occur in a fatigued state. Therefore, new tests need to be developed, or the H:Q ratio may need to be tested under different muscle contraction conditions, in order for relevant, predictive information to be gathered. This proof-of-concept study demonstrates that the hamstring-to-quadriceps$_{Fatigue}$ is different in magnitude and not correlated with the conventional H:Q and, therefore, offers different (new) information that may allow for better injury risk and return-to-sport decisions to be made. However, this test warrants further investigation for its potential to predict hamstring strain and anterior cruciate ligament injury in professional soccer players.

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